Shot on motorola edge 40 neo

2022-26

Full Marks : 70

Time : 3 hours

Answer from **both** the Sections as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

SECTION—A

Answer any *four* questions. 10×4

1. List and explain Registers for the Basic Computer.

2. Draw the neat diagram of decimal adder and explain is functioning by taking suitable example.

3. How do we evaluate the arithmetic statement $X = (A * B) / (C - D)$ using zero, one, two or three address instructions.

4. Write the difference between CISC and RISC. Also, specify the characteristics of both.

5. What is array processor ? Discuss the types and working of array processor.

6. Illustrate how CPU-IOP communication takes place in I/O processor.

7. Define addressing modes. Explain the following addressing modes with examples.

   (*a*) Immediate addressing mode

   (*b*) Indirect addressing mode

   (*c*) Register indirect addressing mode

8. Write an assembly program to print the alphabet *A* to *Z*.

SECTION—B

All questions are compulsory 3×10

9. Define the following terms: *a*. Instruction code *b*. Operation code.

10. How Data transfer takes place from I/O device to CPU in programmed I/O.

11. Write IEEE standard for floating point format.

12. What are the functions of assembler ?

13. What are different types of interrupts available in a computer system ?

14. Define MACRO in assembly program.

15. List any three important features of 8085 processor.

16. Write short notes on Strobe Control method of asynchronous data transfer.

17. What is pipelining ?

18. Add $(356)_8$ and $(173)_8$.

2023(New)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

Answer any **four** questions of the following : 10×4 = 40

1. What are computer instructions ? Draw the instruction code format of basic computer and indicates the number of bits in each part clearly.

2. What do you nunderstand by DMA data transfer ? Give at least one example data transfer situation where DMA mode would be advantages.

3. What do you understand by instruction pipeline ? Mention the typical stages of such a pipeline.

4. What are different addressing modes available for a typical architecture ?

5. What do you understand by asynchronous data transfer ? What are the two methods of asynchronous data transfer ?

6. Write zero address, one address, two address and three anddress instruction code formats for the instruction F = (A * B) – (C / D).

7. Write and assembly program to print even numbers from 1 to 9.

8. What is assembler ? Explain the function of assembler with neat diagram. Differentiate between one pass assembler and two pass assemblers.

Sections – B

Answer **all** qeustions : 3×10 = 30

9. What is the significance of following mnemonics used in assembly program :

   (a) XCHG

(b) INC

(c) DEC

10. What are computer registers ?

11. Define opcode and operand.

12. What is the difference between data and instruction ?

13. Perform addition of 1000 and 1010 using BCD arithmetic.

14. Compare briefly RISC and CISC architecture.

15. What is instructions cycle ?

16. What is program interrupt ?

17. What do you mean by peripheral devices ?

18. Explain birefly floating point representation.

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2022

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

Answer any **four** questions of the following : 10×4 = 40

1. What are the basic difference between Computer Architecture and Computer Organization ?
2. What is asynchronous data transfer ? What are the types of asynchronous data transfer ?
3. What is Interrupt ? Describe different types of interrupt.

4. Define IEEE-754 Standard for floating . Point repersentation with suitable example.

5. Draw a block diagram for data transfer from CPU to an interface.

6. What is DMA ? What are the modes of DMA transfer ?

7. What is the difference between Hardwired control and Micro programmed ?

8. Write a Programme in assembly language to add two single bit Integer.

Sections – B

Answer **all** questions of the following : 3×10 = 30

1. What are the three major phases through which the control units go through an instruction cycle ?

2. Differentiate between RISC and CISC ?

3. Define Opcode. What is the use of Opcode in memory ?

4. Differentiate between Isolated I/O and Memory mapped I/O.

5. Define Register and their types.

6. Define use of Macro in I/C.

7. What are the controls signals used in DMA ?

8. List any three mnemonic code.

9. What is Assembly language ? How it works ?

10. Define Pipeline in computer architecture.

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UGDS(III) — BCA
(2803) Comp. Arch.
& Assem. Lang.

2018

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

Answer any **four** questions.

1. Explain the importance of different addressing modes in computers architecture. 10

2. What are the various registers of 8085 processor ? Discuss their functions. 10

3. What are the three different instruction code formats of a basic computer ? 10

4. Why should we learn assemby language ? Write an assembly program to multiply two numbers. 10

5. Perform the arithmetic operation – 4 *3 with binary numbers in signed – 2's complement represent using Booths algorithm. 10

6. Discuss the phases of instruction cycle in the basic computer. 10

7. Write the mnemonic for following operations in assembly language : 10

   (a) Decrement a register value

   (b) Division operation

   (c) Swapping the values of the registers

   (d) Copying and moving data

   (e) Increment a register value

8. Evaluate arithmetic statement : 10

   $$X = (A * B) + C$$

   Using an :

   (a) General register three address instructions

   (b) General register two address instructions

   (c) Accumulator type one address instructions

   (d) Stack organised computer with zero address operation instructions

Section – B

Answer **all** questions.

9. Compare RISC pipeline vs CISC pipeline. 3

10. Derive the algorithm in flow chart for calculating the square root of a number. 3

11. What do you mean by asynchronous data transfer ? 3

12. What is the difference between vector and array processing ? 3

13. Name three peripheral devices that produce an acceptable output for a person to understand. 3

14. Explain seek and latency time of an external memory. 3

15. What is Priority Interrupt ? 3

16. What do you mean by Input-Output Interface ? Illustrate. 3

17. What are the different modes of data transfer to and from peripherals ? 3

18. Discuss, in brief, control, data and address bus. 3